



Omniverse Compute (OMC)

A Decentralized GPU Computing Network

Whitepaper v3.0

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1. Executive Summary

Omniverse Compute (OMC) is a decentralized physical infrastructure network (DePIN) that connects owners of idle GPU capacity with people who need compute: AI developers, Web3 and decentralized-AI projects, and rendering studios. Providers earn OMC and fees by running verified jobs. Requesters get access to GPU capacity without cloud quotas, long-term contracts or vendor lock-in.

OMC is built on three ideas:

1. **Honest scope.** OMC targets workloads that distribute well across independent machines: inference, fine-tuning on a single node, batch rendering and other embarrassingly parallel jobs. It does not claim to replace tightly coupled data-center clusters for frontier-scale model training.
2. **Verification you can price.** Every job is covered by a verification policy chosen by job type, combining spot-check re-execution, optional full redundancy, hardware-based confidential computing where available, and economic penalties through staking and slashing.
3. **Token demand tied to usage.** Fees are paid per job, a share of every fee is used to buy back and burn OMC, and reward emissions are paid only against verified work.

Settlement and governance run on BNB Chain. High-frequency scheduling, heartbeats and micropayments run on a dedicated Layer 2 rollup built on the OP Stack (opBNB). Jobs execute off-chain in sandboxed containers on provider hardware.

2. Problem and Opportunity

- Demand.** Generative AI has made GPU compute a scarce, expensive input. Small teams and independent developers face high hourly prices, regional quota limits and dependence on a few cloud vendors. Rendering studios face sharp, short-lived peaks in demand.
- Supply.** Large amounts of capability sit underused: consumer and workstation GPUs, idle capacity in small data centers, and hardware formerly used for proof-of-work mining.
- Gap.** There is no widely trusted way for strangers to rent each other GPU time with guarantees about correctness, data protection and payment. OMC aims to provide that missing layer.

Target users

Segment	Typical workloads	What they need
AI startups and independent developers	Inference, LoRA/QLoRA fine-tuning, batch embedding jobs	Lower cost, no quota waits, simple APIs
Web3 and decentralized-AI projects	Agent inference, ZK proof generation	Crypto-native payment, censorship resistance
Rendering and 3D studios	Blender batch rendering, Unreal Engine offline rendering, 3D reconstruction	Burst capacity, per-job pricing
GPU owners and small data centers	Monetizing idle hardware	Predictable earnings, low operating friction

Cost position. We expect savings versus hyperscale on-demand pricing for suitable workloads, but we do not publish a fixed percentage until it is measured. Published benchmarks compare delivered cost per job, including verification overhead, against named alternatives. The benchmark methodology and the underlying rate data are published with the DCF forum *GPU Pricing Watch* series (see Section 13); the first delivered-cost report is published alongside the mainnet release.

3. Supported Workloads and Limits

Workload class	Status	Notes
Batch rendering (Blender, UE offline)	Supported on testnet; mainnet rollout in Phase 2	Deterministic seeds allow tolerance-based verification
Model inference	Supported on testnet; mainnet rollout in Phase 2	Latency-sensitive jobs use region-aware scheduling
Single-node fine-tuning (LoRA/QLoRA)	Supported on testnet; mainnet rollout in Phase 2	Job must fit within one node's VRAM
Multi-GPU jobs within one machine	Supported where the node has the required interconnect	Interconnect such as NVLink is a node attribute, declared and verified at registration
Multi-node distributed training over the public internet	Not supported	Bandwidth and latency make this impractical on consumer networks
Cluster jobs on co-located data-center nodes	Research track	Requires verified low-latency fabric between nodes

Consumer GPUs generally do not offer NVLink. OMC treats interconnect as a declared, tested hardware attribute and never assumes it.

4. Architecture

4.1 Layers

L1 settlement (BNB Chain). OMC token contract, staking contracts, DAO governance, and final dispute arbitration.

L2 rollup orchestration. Node heartbeats, job allocation records, state-channel settlement and micropayments run on an OP Stack rollup on BNB Chain (opBNB). At launch a single sequencer operated by the project submits batches to L1. The published path to decentralization covers rotating sequencer sets, and the rollup's data-availability layer, challenge window, withdrawal delay and user exit procedure are documented in the technical documentation released with mainnet.

Off-chain compute grid. Provider nodes run jobs in isolated containers with CUDA or ROCm acceleration. Inputs and outputs move over encrypted peer-to-peer or relayed channels.

4.2 Job Lifecycle

1. The requester submits a job: container image, resource needs (VRAM, driver version, region), maximum price, verification policy and data-protection level.
2. **Escrow:** the maximum price is locked in the L2 payment contract.
3. The scheduler selects eligible nodes (Section 4.3).
4. The node pulls the image, receives encrypted inputs, executes the job and returns outputs with an execution attestation.
5. Verification runs according to the job's policy (Section 5).
6. **Settlement:** if verification passes, the node is paid from escrow, the protocol fee is split (Section 7.1) and the node's reputation is updated. If it fails, the job is re-run on another node and the faulty node is penalized (Section 7.5).

4.3 Scheduling

Scheduling has two stages.

Stage 1: hard filters. A node is eligible only if it meets the job's minimum VRAM, driver and CUDA/ROCm versions, region constraints, confidential-computing requirement if any, minimum stake and minimum reputation.

Stage 2: scoring. Each eligible node is scored on normalized metrics. Each metric is converted to a percentile rank (0 to 1) among eligible nodes for that job class, so that different units can be combined fairly:

$$\text{Score} = w_f \times N(\text{TFLOPS}) + w_v \times N(\text{VRAM}) + w_b \times N(\text{Bandwidth}) + w_l \times N(\text{Latency, lower is better}) + w_r \times N(\text{Reputation})$$

The weights sum to 1 and are set per job class. **Proposed** defaults:

Job class	TFLOPS	VRAM	Bandwidth	Latency	Reputation
Batch and rendering	0.30	0.20	0.15	0.10	0.25
Inference	0.20	0.15	0.10	0.30	0.25

Assignment is solved per scheduling window as a weighted matching problem that maximizes total score across all pending jobs and eligible nodes. To avoid concentrating work on a few top nodes, the final pick is drawn at random from the top-k scoring candidates. Weights and k are governance parameters.

5. Verification and Trust

No single technique can verify every GPU workload, so OMC uses a policy selected per job.

Mechanism	How it works	Best for	Cost to requester
Spot-check re-execution	A random share of jobs (audit rate a) is silently re-run on an independent node and results are compared	All deterministic or near-deterministic jobs	Base price $\times (1 + a)$. Default a is proposed at 5% to 10%
Full redundancy	The job runs on 2 or 3 independent nodes and outputs are compared	High-value or high-precision jobs, chosen by the requester	2 $\times$ to 3 $\times$
Confidential computing (TEE)	Job runs inside a hardware-protected environment with remote attestation	Sensitive data or proprietary models; requires supporting GPUs	Premium for TEE-capable nodes
Zero-knowledge proofs	Cryptographic proof of execution	Research track; currently too costly for general GPU workloads	Not offered at launch

Handling non-determinism. GPU floating-point results can differ across devices and drivers. OMC therefore compares outputs using task-specific rules: exact hashes for deterministic jobs, numerical tolerance or perceptual metrics for rendering, and fixed seeds with deterministic kernels where inference supports it. Job types that cannot be verified reliably are labeled as such, and requesters are told before submitting.

Reputation. Each node has a score built from verified completions, uptime, timeouts and dispute outcomes. Scores decay without activity and cannot be transferred.

Disputes. A requester may dispute a result within 48 hours of the output being returned (**proposed**). The protocol first re-executes the job on an independent node. If the parties still disagree, the case escalates to L1 arbitration by a DAO-elected arbitrator set drawn from a pool of nine candidates, each posting a bond; a ruling is issued within seven days of escalation (**proposed**). Evidence (attestations, logs, hashes) is recorded on L2.

6. Security and Privacy

Provider hardware is untrusted by default. OMC applies defense in depth, and is explicit about what it cannot guarantee.

Isolation. Jobs run in hardened sandboxes — gVisor by default, with higher-isolation tiers (Kata Containers or microVMs) selectable for sensitive workloads (**proposed**) — with no host access and no outbound network access by default.

Encryption. Inputs, outputs and images are encrypted in transit and at rest. Keys are released to a node only after it is assigned the job.

Data-protection levels. Standard (public or non-sensitive data), Protected (encrypted, ephemeral storage wiped after the job), and Confidential (TEE-only nodes).

Honest limitation. On non-TEE nodes, a malicious provider could in principle observe data or model weights in GPU memory. Requesters with proprietary models or regulated data should choose the Confidential level or not use the network.

Provider protection. Resource limits and abuse detection protect providers from malicious containers. Prohibited content and workloads are defined in the Acceptable Use Policy published on the official website.

Assurance. Smart contracts and the node client are audited before mainnet and the audit reports are public (see Section 11). A bug bounty program runs from mainnet, paying severity-based rewards starting at \$50,000 for critical findings on a public bounty platform (**proposed**).

7. Economics and the OMC Token

Parameters marked **proposed** in this section are design targets: they will be validated on testnet and may change through governance before mainnet.

7.1 Payments and Fee Flows

Requesters pay per job in OMC or in supported stablecoins (USDT and USDC on BNB Chain). Paying in OMC receives a 10% fee discount (**proposed**). A protocol fee of 3% is applied at each job settlement and is split as follows:

- 30% of the fee is used to buy back OMC on the market and burn it automatically through smart contracts.
- 70% goes to the Reputation Reward Pool, held by the Treasury contract and distributed to nodes in proportion to their reputation-weighted verified work.

The burn equals **0.9% of settled job volume** ($3\% \times 30\%$). As an illustration only, not a forecast: if monthly settled volume were \$10 million, the monthly buyback and burn would be about \$90,000. Actual volume, and therefore burn, depends entirely on network usage. The protocol publishes settled volume, fees, burns and emissions on a public dashboard.

7.2 Token Utility

- **Payment** for compute jobs (with a fee discount).
- **Staking** by node operators as security for honest behavior.
- **Governance** voting on protocol parameters, treasury use and upgrades.
- **Rewards** to providers for verified work.

OMC has a fixed maximum supply of 1,000,000,000 tokens on BNB Chain (BEP-20). The only mechanisms that reduce supply are the fee buyback-and-burn and the slashing burn.

7.3 Token Distribution

Category	Share	Amount (OMC)	Vesting
Compute mining and node rewards	50%	500,000,000	0% at TGE. Released over 10 years according to the emission schedule in 7.4
Institutional investors	10%	100,000,000	5% at TGE, 6-month cliff, then 24 months monthly linear
Private investors	8%	80,000,000	10% at TGE, 3-month cliff, then 18 months monthly linear
Community airdrop	2%	20,000,000	100% at TGE
Ecosystem and grants	15%	150,000,000	5% at TGE, 6-month lock, then 36 months monthly linear
Team and core contributors	10%	100,000,000	0% at TGE, 12-month cliff, then 24 months monthly linear

Category	Share	Amount (OMC)	Vesting
Treasury and liquidity	5%	50,000,000	50% at TGE for initial liquidity; the remainder managed by the DAO Treasury

The airdrop is listed separately from investor allocations. Circulating supply at TGE is about 65.5 million OMC (6.55% of maximum supply): 20M airdrop, 5M institutional, 8M private, 7.5M ecosystem and 25M treasury liquidity. A public vesting dashboard shows locked and unlocked balances for every category.

7.4 Emission Schedule for Node Rewards

The 500,000,000 OMC reward pool is released in **equal annual instalments over 10 years**. The ceiling is the same in every year, and within a year release is linear by day. There is no halving step and no epoch structure: a provider that joins in year 7 is mining against the same annual ceiling as one that joined in year 1.

Year	Emission ceiling (OMC)	Share of pool
1	50.00M	10.0%
2	50.00M	10.0%
3	50.00M	10.0%
4	50.00M	10.0%
5	50.00M	10.0%
6	50.00M	10.0%
7	50.00M	10.0%
8	50.00M	10.0%
9	50.00M	10.0%
10	50.00M	10.0%

The schedule is deliberately flat. A halving curve front-loads roughly three quarters of the pool into the first four years and then leaves late-joining providers with a fraction of the early rate; a flat curve keeps the reward rate for a given unit of verified work legible for the whole decade, which is what a provider buying hardware to run for several years actually needs to underwrite. The supply that is not released in the early years is not removed from the pool — it is simply released later, so the total is unchanged at 500,000,000 OMC over 10 years.

Emission is a ceiling, not an entitlement. Rewards are paid only against verified work. Daily emission is scaled by network utilization (**proposed** function: daily emission = ceiling × min(1, U / U_{target}), where U = paid GPU-hours ÷ online available GPU-hours and U_{target} = 60%). Any emission not distributed goes to an Emission Reserve that can only be burned or re-released by DAO vote. Because the annual ceiling is flat, an under-utilized network accumulates a smaller reserve per year than an exponentially decaying schedule would, and that reserve can only re-enter circulation by DAO vote. This ties new supply to real usage rather than to the calendar.

7.5 Staking and Slashing

Nodes must stake OMC before receiving jobs. The required stake is set in USD terms and converted at a time-weighted average price, so that it does not drift with token volatility:

$$\text{Staking_Min (OMC)} = \text{Base_Deposit_USD} \times \text{Tier_Multiplier} \div \text{OMC_TWAP}$$

Tier	Hardware	Tier_Multiplier
1	Consumer GPUs (up to 24 GB VRAM)	1× (proposed)
2	Workstation and professional GPUs	2× (proposed)
3	Data-center GPUs (80 GB VRAM and above)	4× (proposed)

Base_Deposit_USD is \$200 (**proposed**), set before mainnet and adjustable by governance.

The mechanism matters more than the numbers. Because the requirement is denominated in USD and converted at TWAP, the *economic* size of the deposit stays fixed while the OMC amount floats — a provider's exposure does not change just because the token moved. A hard-coded token amount cannot do this: it silently becomes a different economic requirement every time the price moves, which is why the **testnet** ladder (a fixed 20 / 100 / 500 / 1000 / 5000 tOMC) is explicitly a testnet convenience, adopted so that the staking, registration, heartbeat and slashing paths can be exercised with faucet tokens alone. It is not the mainnet rule and is not a template for it.

Slashing triggers and **proposed** penalties (percent of stake):

Offense	Penalty
Job timeout	2%
Unexpected offline during an active job	5%
Incorrect output (failed verification)	30%
Proven fraud or fake heartbeat	100%

Slashed tokens are allocated 50% to compensate affected requesters, 30% to the DAO Treasury and 20% burned (Burn = Slashed × 20%). Penalty sizes will be calibrated on testnet and published before mainnet.

8. Governance

Governance moves from team-led to community-led in stages.

Launch stage. Protocol upgrades are controlled by a 4-of-7 multisignature wallet held by core contributors, with a 3-day timelock (**proposed**). All changes are announced publicly before execution. Individual signers are not disclosed at this stage; the signing policy, key-management practice and any signer rotation will be published before mainnet.

Transition stage. OMC holders vote on parameters (scheduling weights, audit rates, fee split, slashing sizes, emission scaling) and Treasury spending. **Proposed** thresholds: a proposal threshold of 100,000 OMC, a quorum of 10% of circulating supply, a 7-day voting period and a 2-day execution delay.

Full DAO stage (2027 or later). The multisig is removed or limited to emergency security actions with transparent, time-limited powers.

Treasury reports are published quarterly.

9. Competitive Landscape

OMC competes with centralized hyperscale clouds, centralized GPU marketplaces, and other decentralized compute networks. Sources and dates for every figure are listed in Section 13.

Dimension	Hyperscale cloud	GPU marketplaces	Other decentralized networks	OMC
Price per GPU-hour (H100 SXM, on-demand, Aug–Oct 2026)	\$6.88–\$12.29	\$2.49–\$6.16 (Lambda, RunPod, Vultr, CoreWeave)	\$1.25–\$3.00 (Akash, io.net, Render)	Target: below marketplace median; measured benchmark to be published (\$2)
Verification of results	Provider trust	Platform trust	On-chain attestation (Akash); Proof-of-Render and node tiers (Render); proof-of-time-difference and hardware scans (io.net)	Tiered policy per job, see Section 5
Confidential computing	Confidential VMs available; H100 CC on selected instances	Limited	Limited / varies by network	Optional TEE tier
Payment	Fiat	Fiat and crypto	Native token (AKT / IO / RENDER)	OMC and stablecoins
Supported workloads	Broad, including multi-node training	Broad, including multi-node clusters	Mainly single-node inference, fine-tuning and rendering; multi-node training is impractical	See Section 3

Hyperscaler figures are per-GPU-hour for H100 SXM5 80GB in 8-GPU instances (AWS p5.48xlarge \$6.88; Google Cloud a3-highgpu-8g \$11.06; Azure ND96isr H100 v5 \$12.29), spot/preemptible runs \$1.95–\$2.80, with 1-year commitments around –30% to –38% and 3-year around –50% to –62% (vendor published pricing pages, observed September 2026 and cross-checked against independent trackers). Decentralized network figures are ranges published across 2026 sources (decentralised.news, 2026; gpumarketdepin.com, 2026) and vary by supply and workload class. Network uptime reported for decentralized networks (Akash 99.4%, Render 99.8%, io.net 98.6%) remains below hyperscaler SLAs (99.9%–99.99%).

The characteristics OMC intends to differentiate on — verification policy by job type, USD-denominated staking, utilization-linked emissions and BNB Chain ecosystem integration — are stated here as design commitments; measured evidence will be published as the network runs.

10. Roadmap

Phase	Target	Deliverables	Status and evidence
1	Q4 2026 – Q1 2027	L2 contract deployment, open-source node client, testnet, community airdrop	In progress. Testnet live on BNB Chain testnet (Chapel); token contract 0x3c7edae9da38b72db7ae98921ef0759d19de7cc ; explorer testnet.bscscan.com/token/0x3c7edae9da38b72db7ae98921ef0759d19de7cc ; repository github.com/CBTdao/omc-network ; 2% community airdrop (20,000,000 OMC) opens November 1, 2026
2	Q2–Q3 2027	Mainnet launch, reward emission start, Hugging Face and Blender integrations, completed audits	Planned. Mainnet launch is targeted for Q2–Q3 2027; the start of reward emissions, the Hugging Face and Blender integrations and the first completed audits follow within this phase. Audit reports are published here as they are completed
3	2027 Q4 and later	TEE tier, zero-knowledge verification research results, DAO transition, cross-chain settlement	Planned

Each milestone will be accompanied by measurable indicators published on the dashboard: active nodes, total TFLOPS, jobs completed, verified failure rate, settled volume and average time to settlement.

11. Team, Audits and Legal

Core team and advisors. OMC is developed by a pseudonymous collective of core contributors. Consistent with that practice, the project does not publish individual names, employment histories or personal profiles at this stage, and no advisory board is disclosed. All code contributions are public and attributable to the contributing accounts in the repository. If named advisors are engaged, their roles and the scope of their engagement will be published in this section.

Legal entity and jurisdiction. The network and the token are operated as an open-source project. The legal entity that will issue and hold the token, its registration details and the jurisdiction governing the token will be published in this section and on the official website before the token generation event. Until that disclosure is made, no other statement of entity or jurisdiction should be relied upon.

Funding. No external financing has been raised to date; development has been funded by contributor resources. If financing is raised, the amount, the investors and any terms that affect token holders will be disclosed here.

Smart contract audits. Independent audits of the smart contracts and the node client will be completed before mainnet launch. The full reports — auditor, scope, date and the status of every finding — will be published in this section. No audit report has been published yet, and nothing in this document should be read as audit assurance.

Contacts:

- Website: <https://omc.network>
- Forum: <https://forum.omc.network>
- Repository: <https://github.com/CBTdao/omc-network>
- X: https://x.com/Omniverse_Depin
- Telegram: <https://t.me/OmniverseCompute>
- Contact: support@omc.network
- Security disclosure: security@omc.network

12. Risk Factors

Technical risk. Verification methods may fail on some workloads, smart contracts may contain bugs despite audits, and the L2 rollup may have availability or security issues.

Security and privacy risk. Non-TEE nodes cannot fully protect data in use. Sandbox escapes and side-channel attacks are possible.

Market risk. Token prices can be extremely volatile and may fall to zero. Liquidity may be thin. Unlock events may add selling pressure.

Adoption risk. The network needs enough requesters and providers. If demand is low, reward emissions are not matched by fee revenue and the burn is small.

Competition risk. Established clouds and other decentralized networks may offer lower prices or better reliability.

Hardware and supply risk. GPU prices, availability, export controls or energy costs may affect the provider base.

Regulatory risk. Laws on tokens, data protection and cross-border services may change and may restrict use in some countries.

Governance risk. Voting power may concentrate, and early-stage control by a multisig carries key-management risk.

Dependency risk. OMC depends on BNB Chain and third-party infrastructure.

This list is not exhaustive.

13. References and Data Sources

Market and infrastructure data cited in this document:

1. *GPU pricing across providers in 2026 (H100, A100, L4 ranked)* — cloudprice.app, 2026. <https://cloudprice.app/blog/gpu-pricing-across-providers-2026-h100-a100-l4>
2. *Cloud GPU Pricing Comparison 2026: AWS vs Azure vs GCP* — nOps, August 2026. <https://www.nops.io/blog/cloud-gpu-pricing-optimization>
3. *Cloud GPU Pricing Comparison 2026: AWS vs Azure vs GCP* — Economize, 2026. <https://economize.cloud/blog/cloud-gpu-pricing-comparison>
4. *AWS vs Azure vs GCP: H100 GPU Pricing 2026* — Tech Insider, 2026. <https://tech-insider.org/aws-vs-azure-vs-google-cloud-gpu-pricing-2026/>
5. *The Decentralized Compute Risk Matrix: Render vs. Akash vs. io.net* — Decentralised News, 2026. <https://decentralised.news/the-decentralized-compute-risk-matrix>
6. *GPU Market DePIN 2026: Decentralized Compute vs Big Tech* — gpumarketdepin.com, 2026. <https://gpumarketdepin.com/gpu-market-depin-2026-decentralized-compute-vs-big-tech>
7. *Akash vs io.net vs Render: Best DePIN GPU for AI Training* — Misar.Blog, 2026. <https://www.misar.blog/@synor/articles/akash-vs-ionet-vs-render>
8. *Render vs Akash vs io.net: Cheapest DePIN GPU Networks for AI Inference 2026* — Crypto AI World, February 2026. <https://cryptoaiworld.com/2026/02/22/render-vs-akash-vs-io-net-cheapest-depin-gpu-networks-for-ai-inference-2026>

Benchmark methodology for OMC's own delivered-cost claims (Section 2) is published with the DCF forum "GPU Pricing Watch" series: <https://forum.omc.network>

All third-party figures were verified against the sources above at the time of writing. GPU prices change quickly; readers should confirm current rates before relying on them.